

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: Ethical Hacking

COURSE CODE:ITA1407

COURSE OUTCOME COVERED

CO2: Foot printing Tools –: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min Total Marks: 50

Scenario-Based Assessment

Q1 – Footprinting Tools: Conducting Competitive Intelligence

Step 1 – Domain Registration Details

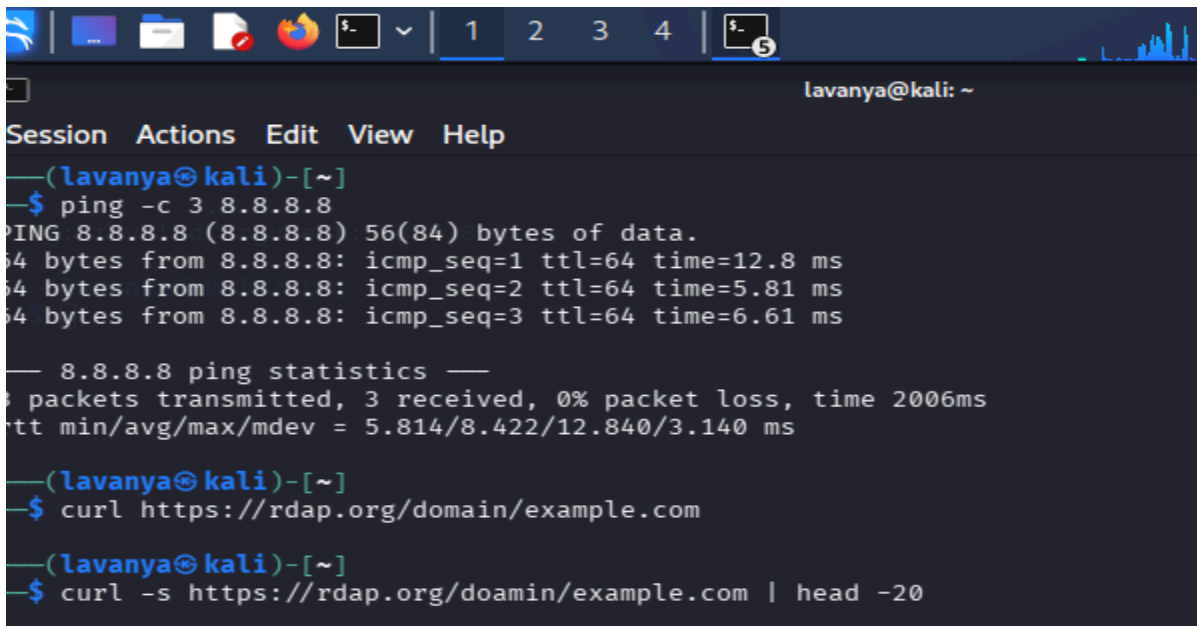
Command:

whois example.com

If you get **Connection refused**, use this instead:

curl -s https://rdap.org/domain/example.com | head -30

What it demonstrates: Domain registration and DNS-related information.

A screenshot of a Kali Linux terminal window. The window has a dark background with a menu bar at the top containing 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows the user 'lavanya@kali' in the prompt. The first command executed is 'ping -c 3 8.8.8.8', which returns three lines of ping data and a summary: 'PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.', '64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=12.8 ms', '64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=5.81 ms', '64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=6.61 ms', followed by '— 8.8.8.8 ping statistics —', '3 packets transmitted, 3 received, 0% packet loss, time 2006ms', and 'rtt min/avg/max/mdev = 5.814/8.422/12.840/3.140 ms'. The second command is 'curl https://rdap.org/domain/example.com'. The third command is 'curl -s https://rdap.org/doamin/example.com | head -20'. The window title bar shows several application icons and a tab labeled '5'.

Step 2 – Find Public Email Addresses

Command:

theHarvester -d example.com -b bing

If **theHarvester** is not installed:

sudo apt update

sudo apt install theharvester

Then run:

theHarvester -d example.com -b bing

What it demonstrates: Collection of publicly available email addresses and hostnames.

```
(lavanya@kali)-[~]
$ theHarvester -d example.com -b bing
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*                                                                 *
*  _ _ _ _ _ ^ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ *
*  | _ | _ | \ / _ | \ / _ | \ / _ | \ / _ | \ / _ | \ / _ | *
*  | _ | _ | \ / _ | \ / _ | \ / _ | \ / _ | \ / _ | \ / _ | *
*  \ _ | _ | \ / _ | \ / _ | \ / _ | \ / _ | \ / _ | \ / _ | *
*                                                                 *
* theHarvester 4.11.1                                           *
* Coded by Christian Martorella                                 *
* Edge-Security Research                                        *
* cmartorella@edge-security.com                                *
*                                                                 *
*****
The following engines are not supported: {'bing'}

[!] Invalid source.
```

Step 3 – Identify Technology Stack

Command:

whatweb https://example.com

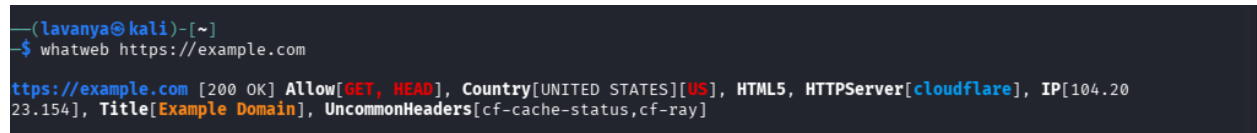
If **whatweb** is not installed:

sudo apt install whatweb

Then:

whatweb https://example.com

What it demonstrates: Identification of publicly observable web technologies.



```
—(lavanya@kali)-[~]  
$ whatweb https://example.com  
  
https://example.com [200 OK] Allow[GET, HEAD], Country[UNITED STATES][US], HTML5, HTTPServer[cloudflare], IP[104.20  
23.154], Title[Example Domain], UncommonHeaders[cf-cache-status,cf-ray]
```

Step 4 – Find Public Subdomains

Command:

subfinder -d example.com

If **subfinder** is not installed:

sudo apt install subfinder

Then:

```
subfinder -d example.com
```

What it demonstrates: Discovery of publicly known subdomains.

```
(lavanya@kali)-[~]  
$ subfinder -d example.com  
  
projectdiscovery.io  
  
[INF] Current subfinder version v2.14.0 (outdated)  
[INF] Loading provider config from /home/lavanya/.config/subfinder/provider-config.yaml  
[INF] Enumerating subdomains for example.com
```

Step 5 – Identify DNS Nameservers

Command:

```
dig NS example.com
```

What it demonstrates: Identification of authoritative DNS nameservers.

Q2 – DNS Zone Transfers

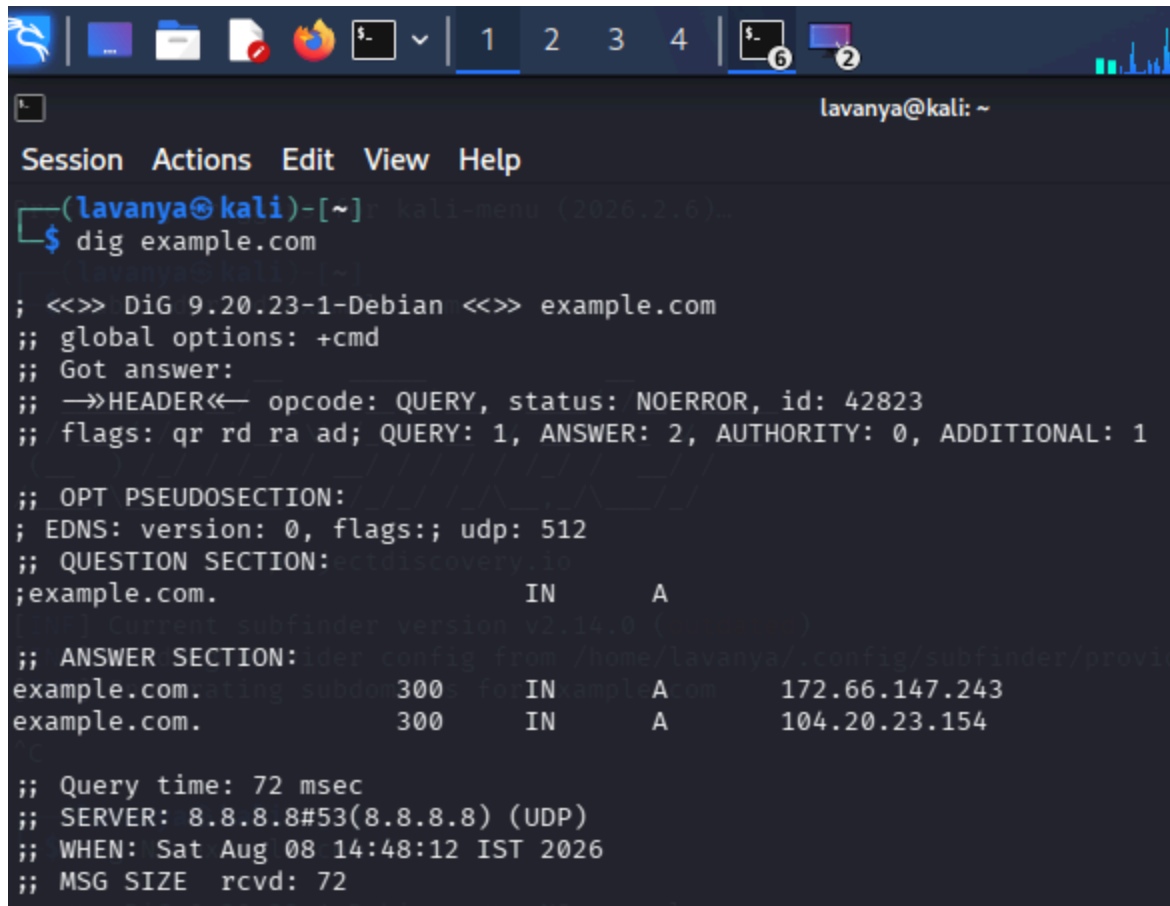
Step 1 – Check DNS Resolution

Command:

```
dig example.com
```

What it demonstrates:

Checks whether the domain can be resolved through DNS and displays its DNS information.



```
lavanya@kali: ~  
Session Actions Edit View Help  
(lavanya@kali)-[~]  
$ dig example.com  
; <<>> DiG 9.20.23-1-Debian <<>> example.com  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 42823  
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:;; udp: 512  
;; QUESTION SECTION:  
;example.com. IN A  
  
;; ANSWER SECTION:  
example.com. 300 IN A 172.66.147.243  
example.com. 300 IN A 104.20.23.154  
  
;; Query time: 72 msec  
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)  
;; WHEN: Sat Aug 08 14:48:12 IST 2026  
;; MSG SIZE rcvd: 72
```

Step 2 – Find Authoritative DNS Nameservers

Command:

dig NS example.com

What it demonstrates:

Identifies the authoritative DNS nameservers responsible for the domain.

```
(lavanya@kali)-[~]
$ dig NS example.com

; <<>> DiG 9.20.23-1-Debian <<>> NS example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 54694
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512      NS      elliottns.cloudflare.com.
;; QUESTION SECTION:
;example.com.                IN      NS      herans.cloudflare.com.

;; ANSWER SECTION:
example.com. 8.8.8.8 IN NS      elliottns.cloudflare.com.
example.com. 8.8.8.8 IN NS      herans.cloudflare.com.

;; Query time: 76 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Aug 08 14:48:26 IST 2026
;; MSG SIZE rcvd: 95
```

Step 3 – Perform DNS Zone Transfer Test

Command:

dig AXFR example.com @a.iana-servers.net

Note: Replace [a.iana-servers.net](#) with the authoritative nameserver obtained from Step 2.

What it demonstrates:

Tests whether the DNS server allows an AXFR (zone transfer) request.

```
(lavanya@kali)-[~]
$ dig AXFR example.com @a.iana-servers.net

; <<>> DiG 9.20.23-1-Debian <<>> AXFR example.com @a.iana-servers.net
;; global options: +cmd
; Transfer failed.
```

Step 4 – Test the Second DNS Server

Command:

dig AXFR example.com @b.iana-servers.net

Note: Replace b.iana-servers.net with the second authoritative nameserver obtained from Step 2.

```
(lavanya@kali)-[~]  
$ dig AXFR example.com @b.iana-servers.net  
; QUESTION SECTION:  
; <<>> DiG 9.20.23-1-Debian <<>> AXFR example.com @b.iana-servers.net  
;; global options: +cmd  
; Transfer failed.
```

Step 5 – Identify Mail Servers

Command:

dig MX example.com

What it demonstrates:

Identifies the mail servers configured for the domain.

```
(lavanya@kali)-[~]  
$ dig MX example.com  
dig ;; communications error to 8.8.8.8#53: timed out  
A example.com  
;; communications error to 8.8.8.8#53: timed out  
;; communications error to 8.8.8.8#53: timed out  
; flags: qr rd ra  
; <<>> DiG 9.20.23-1-Debian <<>> MX example.com  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: FORMERR, id: 24860  
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1  
;; WARNING: recursion requested but not available  
; ANSWER SECTION:  
;; OPT PSEUDOSECTION: 21600 IN NS elliott.ns.cloudflare.com  
;; EDNS: version: 0, flags:;; udp: 1232 NS hera.ns.cloudflare.com  
;; COOKIE: 2f6ca52692c4ea67 (echoed)  
;; QUESTION SECTION:  
; example.com. 8.8.8.8#53(8.8.8.8) IN MX  
; WHEN: Sat Aug 08 14:37:04 IST 2026  
;; Query time: 96 msec  
;; SERVER: 172.21.100.31#53(172.21.100.31) (UDP)  
;; WHEN: Sat Aug 08 14:50:04 IST 2026  
;; MSG SIZE rcvd: 52
```

Step 6 – Identify IP Address

Command:

dig A example.com

What it demonstrates:

Identifies the IPv4 address associated with the domain.

```
(lavanya@kali)-[~]
$ dig A example.com
; Flags: qr rd ra --
; <<> DiG 9.20.23-1-Debian <<> A example.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 29732
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com. IN A
; Query time: 76 msec
;; ANSWER SECTION:
example.com. Aug 08 14:30:04 IST IN A 172.66.147.243
example.com. rcvd: 95 300 IN A 104.20.23.154

;; Query time: 72 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Aug 08 14:50:04 IST 2026
;; MSG SIZE rcvd: 72
```

Step 7 – Display DNS Records

Command:

dig ANY example.com

What it demonstrates:

Attempts to display available DNS record information such as A, MX, NS, and other records. Some DNS servers may refuse or limit **ANY** queries.

```

(lavanya@kali)-[~]
$ dig ANY example.com

; <<>> DiG 9.20.23-1-Debian <<>> ANY example.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 43837
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com. 3600 IN ANY
;example.com. 3600 IN ANY

;; ANSWER SECTION:
example.com. 3600 IN HINFO "RFC8482" ""
example.com. 3600 IN RRSIG HINFO 13 2 3600 20260809102019 20260807082019 3450
KPCow3IvELd4G8taFd+lZ//Vuxfyp3QjuAaubYt0kjV0sU1M TTLxk3yAPmudKPyIyT4blyVX9hlftA=

;; Query time: 8 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (TCP)
;; WHEN: Sat Aug 08 14:55:56 IST 2026
;; MSG SIZE rcvd: 168

```

Q3 – Introduction to Social Engineering: Social Engineering Attacks and Countermeasures

Step 1 – Create a Sample Phishing Email

Command:

nano phishing_email.txt

Enter the following sample email:

From: IT Department <it-support@example.com>

Subject: URGENT - Verify Your Account

Your account requires urgent security verification.

Click the link below to verify your login credentials:

`http://fake-login.example.net/verify`

Failure to verify your account may result in account suspension.

Save the file using:

Ctrl + O → Enter → Ctrl + X

What it demonstrates:

Creates a sample phishing email for security analysis without sending an actual email.

Step 2 – Display the Email

Command:

`cat phishing_email.txt`

What it demonstrates:

Displays the contents of the suspicious email for investigation.

```
(lavanya@kali)-[~]  
$ cat phishing_email.txt  
From: IT Department<it-support@example.com>  
Subject: URGENT - Verify Your Account requires urgent security verification.  
  
Click the link below to verify your login credentials:  
  
http://fake-login.example.net/verify
```

Step 3 – Search for Urgent or Suspicious Words

Command:

```
grep -Ein "urgent|verify|login|credential|suspension" phishing_email.txt
```

What it demonstrates:

Identifies common phishing indicators such as urgency, login requests, credential requests, and threats of account suspension.

```
(lavanya@kali)-[~]  
$ grep -Ein "urgent|verify|login|credential|suspension" phishing_email.txt  
2:subject: URGENT - Verify Your Account requires urgent security verification.  
4:Click the link below to verify your login credentials:  
5:http://fake-login.example.net/verify  
8:Failure to verify your account may result in account suspension.
```

Step 4 – Extract the Suspicious Link

Command:

```
grep -Eo 'https?://[^\ ]+' phishing_email.txt
```

What it demonstrates:

Extracts URLs from the email so that suspicious links can be identified before employees click them.

```
(lavanya@kali)-[~]  
$ grep -Eo 'https?://[^\ ]+' phishing_email.txt  
http://fake-login.example.net/verify
```

Step 5 – Check the Sender Information

Command:

```
grep -i "From:" phishing_email.txt
```

What it demonstrates:

Displays the sender address so that employees or analysts can check whether the sender is legitimate.

```
(lavanya@kali)-[~]  
$ grep -i "From:" phishing_email.txt  
From: IT Department<it-support@example.com>
```

Step 6 – Search for Credential Requests

Command:

`grep -Ein "password|username|login|credential" phishing_email.txt`

What it demonstrates:

Detects requests for sensitive login information, which is a major warning sign of credential-phishing attacks.

```
(lavanya@kali)-[~]  
$ grep -Ein "password|username|login|credentials" phishing_email.txt  
grep: phishing_email.txt: No such file or directory  
  
(lavanya@kali)-[~]  
$ pwd  
/home/lavanya  
  
(lavanya@kali)-[~]  
$ ls  
300      Documents  file2.txt  fruits.txt  Music      Projects   report.xml  Videos  
copy.txt Downloads  file3.txt  index.html  nmap-report.txt  Public     Templates  
Demo     EXNO_1.png file.txt   lab         phishing_email.txt  report.gnmap  Test  
Desktop  file1.txt  file.txt.save  linuxlab   Pictures     report.nmap  TestBackup  
  
(lavanya@kali)-[~]  
$ ls phishing_email.txt  
phishing_email.txt
```

